Faheem Shah Umer Vattam Kandathil

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WORK EXPERIENCE

Werkstudent – Applied AI Engineer Sep 2024 – Present
IAV GmbH *Ingolstadt, Germany*

- Developed three production AI applications used by an estimated 50+ users, including a standalone internal RAG assistant that retrieves project information from PDFs, OneNote, emails, Word, Confluence, and Excel.
- Built and evaluated a four-agent ECU analysis system with orchestration, numerical analysis, plotting, and RAG retrieval from documentation and e-handbooks; used Langfuse to trace and assess multi-agent workflows.
- Designed diagnostic logic for fuel tank leakage detection as part of functional safety development.
- Measured and analysed ECU signals using ETAS MDA and INCA, validating ECU safety logic through simulation, numerical signal analysis, and prototype vehicle testing for an automotive OEM.

Masterarbeit Apr 2026 – Oct 2026
A Framework for Large Language Model-Driven Agent-Based Tolerance Management
KTmfk - Lehrstuhl für Konstruktionstechnik | FAU *Erlangen, Germany*

- Developed and evaluated a LangGraph-based framework orchestrating seven specialized agents for tolerance management in product development.
- Validated the prototype on several CAD files while integrating engineering knowledge retrieval, CAD/STEP and PMI processing, Monte Carlo tolerance analysis, visualization, FE meshing, and Siemens VisVSA execution.
- Implemented RAG, structured engineering artifacts, deterministic Python calculations, traceable agent handoffs, and human-in-the-loop clarification.

Projektarbeit Apr 2025 – Sep 2025
Potential Study for the Use of Large Language Models in Tolerance Management *Grade: 1.0*
KTmfk - Lehrstuhl für Konstruktionstechnik | FAU *Erlangen, Germany*

- Built a RAG knowledge base from approximately 4,000 technical documents, including books, articles, standards, project papers, and theses.
- Evaluated the prototype on 20+ engineering questions, achieving approximately 95% retrieval and answer accuracy while measuring response time and user satisfaction.

Software Engineer and Content Creator Jan 2023 – Apr 2023
Baabte System Technologies Pvt Ltd *Calicut, India*

- Built working models and created technical content for Robotics, IoT, and AI projects using Arduino, MicroPython, and Raspberry Pi.
- Tools: Jira, Arduino IDE, Fritzing, MicroPython, NodeMCU, ESP32/ESP8266.

Digital Twin Benchmarking in Electronics Apr 2022 – Jul 2022
Caresoft Global *Coimbatore, India*

- Benchmarked IC components, researched architecture, manufacturer, and production cost, and compiled a final report on an electric vehicle module for use in future upgrades.

Intern Feb 2021 – Mar 2021
Technologics Global Pvt. Ltd *Bangalore, India*

- Trained in ML algorithms and Python using TensorFlow, Keras, and Pandas in Jupyter Notebook.

EDUCATION

Master's in Electromobility ACES | FAU Erlangen Nuremberg | Present | Current GPA: 1.6

- Specializing in AI, autonomous driving, and sustainable mobility.
- Strong knowledge of electromobility technologies, including electronics, AI, and machine learning.
- Experience with simulation and modelling of electromobility systems.
- Proficient in Python.

EDUCATION

Bachelor in Mechatronics Engineering | HICET Coimbatore | Jul 2018 – May 2022 | CGPA: 8.2/10

- Experience with Mechatronics systems, including sensors, actuators, and controllers.
- Proficient in Python, C++, MATLAB, MSC ADAMS, Arduino and Windows Office.

SKILLS

AI / Agentic Systems:	LLMs, RAG, Multi-Agent Systems, LLMOps, Langfuse, LangChain, LangGraph, Hugging Face, OpenRouter, Qdrant, Embeddings, Automatic Speech Recognition, Text-to-Speech, Human-in-the-Loop Workflows, MCP, A2A, Google ADK, SpaCy, NLTK, TensorFlow, Keras.
Engineering / Simulation:	Tolerance Management, CAD/STEP and PMI Processing, Tolerance Analysis, OpenCascade/XCAF, Gmsh, Siemens VisVSA, Matplotlib, MATLAB, MSC ADAMS.
Automotive / Embedded:	Functional Safety (ECU), ETAS MDA, ETAS INCA, Digital Twin, Arduino, MicroPython, Raspberry Pi, Soldering, Welding.
Programming / Tools:	Python, C, C++, PyQt5, PySide6, Streamlit, OpenAI Codex, Claude Code, Git, GitLab, Bitbucket, Jira, Confluence, WordPress, Excel, Word.

PROJECTS

- **SpeakRAG – Voice-Enabled RAG Assistant**
  Built a Python desktop assistant for voice-based PDF Q&A using local automatic speech recognition (ASR) via faster-whisper, cited Qdrant retrieval, OpenRouter answer generation, and Piper text-to-speech (TTS). Made each pipeline stage inspectable and PDF indexing reusable for other documents.
  Tools: Python, PySide6, PyAudio, librosa, faster-whisper, Qdrant, OpenRouter, Piper
  Link: github.com/faheem-shah-umer/SpeakRAG
- **LanternLM – Offline Desktop AI Assistant**
  Built a portable Windows app for fully offline GGUF inference via llama.cpp with streaming, model discovery, SQLite chat history, and a local file workspace in a context-isolated Electron architecture.
  Tools: Electron, React, TypeScript, llama.cpp, GGUF, SQLite, Vite
  Link: github.com/faheem-shah-umer/LanternLM
- **STEP-RAG – Conversational Retrieval for 3D CAD**
  Built a CAD-aware RAG pipeline that parses STEP files, extracts geometry and spatial relationships, and indexes structured summaries in Qdrant; added conversational retrieval, citations, relevance scoring, and a Streamlit interface.
  Tools: Python, OpenCascade, LangGraph, LangChain, Qdrant, Hugging Face, OpenRouter, Streamlit
  Link: github.com/faheem-shah-umer/Step-RAG-chatbot
- **SITY – Agile Workspace Presence Monitor**
  Built a privacy-preserving occupancy monitor for six rooms in a Scrum team using sprint backlogs, iterative prototypes, and reviews; led hardware selection, schematic design, and microcontroller programming.
  Tools: GitLab, Arduino IDE, Firebase, Flask, Miro
  Link: mad.tf.fau.de/research/projects/sity/
- **Portable Robotic Arm (Android-Controlled)**
  Built a sub-3 kg Android-controlled robotic arm and led a four-member team through design, implementation, and documentation.
  Tools: Arduino IDE/SDK, Android Development Tools, C++, Java.

CERTIFICATIONS & TRAINING

Microsoft: Build Cross-Platform Multi-Agent Solutions Using the Agent2Agent Protocol in Microsoft Copilot Studio
LangChain Academy: Foundation: Introduction to LangChain – Python
Foundation: Introduction to LangGraph – Python
Project: Ambient Agents with LangGraph
Project: Deep Research with LangGraph
Google: Deploy Multi-Agent Architectures
University of Toronto / Coursera: Introduction to Self-Driving Cars (August 2020)

LANGUAGES

German: A2 level English: C1 level